

NUM4 – NUM4 TASK 1: DATA CLEANING

DATA CLEANING – D206

PRFA – NUM4

[Task Overview](#)[Submissions](#)[Evaluation Report](#)

COMPETENCIES

4030.3.1 : Predicting Obstacles in Data Analysis

The graduate predicts potential obstacles in data analysis based on the quality of data provided.

4030.3.2 : Preparing Data for Analysis

The graduate prepares data for analysis to address organizational needs.

4030.3.3 : Manipulating Data for Analysis

The graduate writes reusable code to manipulate and clean data in preparation for analysis.

INTRODUCTION

In a previous course, you used Structured Query Language (SQL) methods to collect data for analysis and to support decision-making processes. The next step involves preparing the data for analysis, a process known as data cleaning. You will explore various graphs and statistics to identify outliers, consider various methods to handle missing data such as imputation, and explore a basic use of principal component analysis (PCA) for data reduction of a set of variables.

To complete this assessment, you will use raw data from the industry of your choice and prepare the data set for analysis. You will also create visualizations and deliver a clean data set ready for exploratory analysis.

REQUIREMENTS

Your submission must be your original work. No more than a combined total of 30% of the submission and no more than a 10% match to any one individual source can be directly quoted or closely paraphrased from sources, even if cited correctly. The similarity report that is provided when you submit your task can be used as a guide.

You must use the rubric to direct the creation of your submission because it provides detailed criteria that will be used to evaluate your work. Each requirement below may be evaluated by more than one rubric aspect. The rubric aspect titles may contain hyperlinks to relevant portions of the course.

*Tasks may **not** be submitted as cloud links, such as links to Google Docs, Google Slides, OneDrive, etc., unless specified in the task requirements. All other submissions must be file types that are uploaded and submitted as attachments (e.g., .docx, .pdf, .ppt).*

Note: All visualizations are created with Python or R and should be embedded in the report. Do not use CAD programs because attachments will be too large.

Select **one** of the data files and its associated dictionary file from the "[D206 Definitions and Data Files](#)" web link, then do the following:

Part I: Research Question

Note: Your responses to the task prompts must be provided in a document file. Unless otherwise specified, responses to PA requirements that are included in a Python or R studio notebook will not be accepted.

A. In a document file, describe your research problem by doing the following:

1. Describe **one** question or decision that could be addressed using the data set you chose. The summarized question or decision must be relevant to a realistic organizational need or situation.
2. Describe *all* variables in the data set (regardless of the research question) and indicate the data type for *each* variable. Use examples from the data set to support your claims.

Part II: Data-Cleaning Plan

Note: You may use Python or R for implementing your coding solutions, manipulating the data, and creating visual representations. However, your responses to the task prompts must be provided in a document file. Unless otherwise specified, responses to PA requirements that are included in a python or R studio notebook will not be accepted.

B. In a document file, explain the plan for cleaning the data by doing the following:

1. Propose a plan that includes the relevant techniques and specific steps needed to assess the quality of the data in the data set.
2. Justify your approach for assessing the quality of the data, including the following:
 - characteristics of the data being assessed
 - the approach used to assess the quality of the data
3. Justify your selected programming language and any libraries and packages that will support the data-cleaning process.

Part III: Data Cleaning

Note: Your responses to the task prompts must be provided in a document file. Unless otherwise specified, responses to PA requirements that are included in a Python or R studio notebook will not be accepted.

C. In a document file, summarize the data-cleaning process by doing the following:

1. Describe the findings for the data quality issues found from the implementation of the data-cleaning plan from Part II.
2. Justify your methods for mitigating the data quality issues in the data set.
3. Summarize the outcome from the implementation of *each* data-cleaning step.
4. Summarize the limitations of the data-cleaning process.
5. Discuss how the limitations summarized in part C4 could affect the question analysis or decision from part A1.

- D. Provide the detection and mitigated code you used for the data-cleaning process by doing the following:
1. Provide the annotated code you will use to mitigate the data quality issues—including anomalies—in the data set in an executable script file such as Python or R.
 2. Provide a copy of the cleaned data set as a CSV file.
- E. Apply principal component analysis (PCA) to identify the significant features of the data set and provide the following in a document file:
1. Identify the total number of principal components and provide the output of the principal components loading matrix.
 2. Justify the reduced number of the principal components and include a screenshot of a scree plot.
 3. Describe how the organization would benefit from the use of PCA.

Part IV: Supporting Documents

- F. Provide a Panopto video recording that includes the presenter and a vocalized demonstration of the functionality of the code used for the analysis of the programming environment.

Note: For instructions on how to access and use Panopto, use the "Panopto How-To Videos" web link provided below. To access Panopto's website, navigate to the web link titled "Panopto Access" and then choose to log in using the "WGU" option. If prompted, log in using your WGU student portal credentials, and then it will forward you to Panopto's website.

To submit your recording, upload it to the Panopto drop box titled "[Data Cleaning NUMx | D206 \(Student Creators\) \[assignments\]](#)." Once the recording has been uploaded and processed in Panopto's system, retrieve the URL of the recording from Panopto and copy and paste it into the Links option. Upload the remaining task requirements using the Attachments option.

- G. Acknowledge web sources, using in-text citations and references, for segments of third-party code used to support the application. Be sure the web sources are reliable.
- H. Acknowledge sources, using in-text citations and references, for content that is quoted, paraphrased, or summarized.
- I. Demonstrate professional communication in the content and presentation of your submission.

File Restrictions

File name may contain only letters, numbers, spaces, and these symbols: ! - _ . * ' ()

File size limit: 200 MB

File types allowed: doc, docx, rtf, xls, xlsx, ppt, pptx, odt, pdf, csv, txt, qt, mov, mpg, avi, mp3, wav, mp4, wma, flv, asf, mpeg, wmv, m4v, svg, tif, tiff, jpeg, jpg, gif, png, zip, rar, tar, 7z

RUBRIC

A1: QUESTION OR DECISION

NOT EVIDENT

APPROACHING
COMPETENCE

COMPETENT

A description is not provided in a document file.

The description in the document file does not summarize a question or decision that can be addressed through analysis of the chosen data set. Or the question or decision is not relevant to a realistic organizational need or situation. Or the question does not use *all* variables in the data set.

The description in the document file summarizes a question or decision that can be addressed through analysis of the chosen data set. The question or decision is relevant to a realistic organizational need or situation and uses *all* variables in the data set.

A2:REQUIRED VARIABLES

NOT EVIDENT

A description of the data set is not provided in a document file.

APPROACHING COMPETENCE

The description of the data set in the document file does not include *all* of the variables in the data set or does not indicate the specific type of data being described for *each* variable. Or the description does not include examples from the data set to support claims.

COMPETENT

The description in the document file includes *all* variables in the data set and indicates the specific type of data being described for *each* variable. The description includes examples from the data set to support claims.

B1:PLAN TO ASSESS QUALITY OF DATA

NOT EVIDENT

A proposal is not provided in a document file.

APPROACHING COMPETENCE

The proposal in the document file includes techniques and steps needed for assessing the quality of the data, but the techniques or the steps are not relevant or not appropriate for assessing the quality of the data in the data set.

COMPETENT

The proposal in the document file includes a detailed description of the techniques and steps needed for assessing the quality of the data in the selected data set.

B2:JUSTIFICATION OF APPROACH

NOT EVIDENT

A justification is not provided in a document file.

APPROACHING COMPETENCE

The justification in the document file includes the character-

COMPETENT

The justification in the document file includes the characteristics of the data being assessed, refer-

istics of the data being assessed, but the justification does not reference the approach used to assess the quality of the data. Or the justified approach does not align with the selected data set.

ences the approach used to assess the quality of the data, and aligns with the selected data set.

B3:JUSTIFICATION OF TOOLS

NOT EVIDENT

A justification is not provided in a document file.

APPROACHING COMPETENCE

The justification in the document file describes the benefits of using the programming language, including *any* libraries and packages used to clean the data, but does not include specific examples of how these tools are ideal for cleaning this data as opposed to other available tools.

COMPETENT

The justification in the document file describes the benefits of using the programming language, including *any* libraries and packages used to clean the data and includes specific examples of how these tools are ideal for cleaning this data as opposed to other available tools.

C1:CLEANING FINDINGS

NOT EVIDENT

A description is not provided in a document file.

APPROACHING COMPETENCE

The description in the document file includes the findings for some of the data quality issues found by running the code from Part II, but the description contains inaccuracies, or the description does not include *all* of the data quality issues.

COMPETENT

The description in the document file accurately includes *all* of the data quality issues found by running the code from Part II.

C2:JUSTIFICATION OF MITIGATION METHODS

NOT EVIDENT

A justification is not provided in a document file.

APPROACHING COMPETENCE

The justification in the document file includes mitigation methods that are not specific to the data quality issues listed in part D1. Or the methods could

COMPETENT

The justification in the document file includes the specific mitigation methods for *all* data quality issues listed in part D1.

not be used to mitigate the data quality issues.

C3:SUMMARY OF THE OUTCOMES**NOT EVIDENT**

A summary is not provided in a document file.

APPROACHING COMPETENCE

The summary in the document file details the outcome of the implementation of some but not *all* of the data-cleaning steps. Or the summarized outcomes are not plausible given the interventions.

COMPETENT

The summary in the document file details the outcome of the implementation of *each* data-cleaning step. The summarized expected outcomes are plausible given the interventions.

C4:LIMITATIONS**NOT EVIDENT**

No limitations are summarized, or the summary is not provided in a document file.

APPROACHING COMPETENCE

1 or more relevant limitations of the data-cleaning process are not included in the summary in the document file, or the summary is inaccurate.

COMPETENT

The limitations of the data-cleaning process are summarized completely and accurately in a document file.

C5:IMPACT OF LIMITATIONS**NOT EVIDENT**

The discussion does not address the impact of *any* limitations summarized in part C4 or is not provided in a document file.

APPROACHING COMPETENCE

The discussion in the document file addresses the impact of the limitations summarized in part C4, but the discussion does not align with the question or decision described in part A1.

COMPETENT

The discussion in the document file addresses the impact of the limitations summarized in part C4 and aligns with the question or decision described in part A1.

D1:MITIGATION CODE**NOT EVIDENT**

No mitigated code is provided.

APPROACHING COMPETENCE

The provided mitigated code is incomplete or is not in an annotated, executable script file. Or the code provided could not be

COMPETENT

The provided mitigated code is in an annotated executable script file that could be used to mitigate the data quality issues in the data set.

used to mitigate data quality issues in the data set.

D2:CLEAN DATA**NOT EVIDENT**

No data set is provided.

APPROACHING COMPETENCE

The provided data set contains data quality issues that should have been mitigated. Or the data set is missing variables from the chosen data set. Or the file is not in a CSV format.

COMPETENT

The cleaned data set is provided in a CSV format created from the raw data and includes the complete list of variables from the chosen data set.

E1:PRINCIPAL COMPONENTS**NOT EVIDENT**

Principal components are not identified, or the output of the principal components loading matrix is not provided. Or the submission is not provided in a document file.

APPROACHING COMPETENCE

The submission is provided in the document file, but the total number of identified principal components is inaccurate, or the output of the principal components loading matrix is inaccurate.

COMPETENT

The total number of principal components is accurately identified, and the output of the principal components loading matrix is accurate. The submission is provided in the document file.

E2:CRITERIA USED**NOT EVIDENT**

A justification is not provided in a document file.

APPROACHING COMPETENCE

The justification in the document file for the reduced number of principal components is inaccurate or incomplete. Or a screenshot of a scree plot is not included.

COMPETENT

The justification in the document file for the reduced number of principal components is accurate and complete and includes a screenshot of a scree plot.

E3:BENEFITS**NOT EVIDENT**

A description is not provided in a document file.

APPROACHING COMPETENCE

The description in the document file of how the organization would benefit from PCA is illogical or inaccurate.

COMPETENT

The description in the document file of how the organization would benefit from PCA is logical and accurate.

F:VIDEO

NOT EVIDENT

The submission does not provide a Panopto video recording.

APPROACHING COMPETENCE

A Panopto video recording is provided that includes a demonstration of the functionality of the code used for the analysis, but an identification of the version of the programming environment is not provided, *all* of the listed points are not discussed, or the video does not capture *both* the presenter and the vocalized presentation of the functioning code in a Panopto video recording.

COMPETENT

A Panopto video recording is provided that includes a demonstration of the functionality of the code used for the analysis and an identification of the version of the programming environment. The demonstration includes *all* of the listed points. For the duration of the presentation, the video captures *both* the presenter and the vocalized presentation of the functioning code in a Panopto video recording.

G:SOURCES OF THIRD-PARTY CODE

NOT EVIDENT

Web sources used are not acknowledged.

APPROACHING COMPETENCE

The submission acknowledges only some of the web sources used to acquire data or third-party code. Or the web sources are not reliable.

COMPETENT

The submission acknowledges *all* web sources used to acquire data or third-party code, and *all* of the web sources are reliable.

H:SOURCES

NOT EVIDENT

The submission does not include both in-text citations and a reference list for sources that are quoted, paraphrased, or summarized.

APPROACHING COMPETENCE

The submission includes in-text citations for sources that are quoted, paraphrased, or summarized and a reference list; however, the citations or reference list is incomplete or inaccurate.

COMPETENT

The submission includes in-text citations for sources that are properly quoted, paraphrased, or summarized and a reference list that accurately identifies the author, date, title, and source location as available.

I:PROFESSIONAL COMMUNICATION

NOT EVIDENT**APPROACHING COMPETENCE****COMPETENT**

This submission includes professional communication errors related to spelling, grammar, punctuation, and sentence fluency. For best results, please focus on the specific Correctness errors identified by Grammarly for Education to help guide your revisions. If you need additional assistance preparing your submission, please contact your Instructor.

This submission includes professional communication errors related to spelling, grammar, punctuation, and/or sentence fluency. For best results, please focus on the specific Correctness errors identified by Grammarly for Education to help guide your revisions.

This submission demonstrates correct use of spelling, grammar, punctuation, and sentence fluency. You have demonstrated quality professional communication skills in this submission.

WEB LINKS

[D206 Definitions and Data Files](#)

[Panopto Access](#)

Sign in using the "WGU" option. If prompted, log in with your WGU student portal credentials, which should forward you to Panopto's website. If you have any problems accessing Panopto, please contact Assessment Services at assessmentservices@wgu.edu. It may take up to two business days to receive your WGU Panopto recording permissions once you have begun the course.

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[Panopto How-To Videos](#)